US-based aWhere is one such company that gives smart climate insights using machine learning algorithms to farmers, crop consultants and retailers.

Soil parameters measured using sensors are texture, salinity levels, nutrient status, organic content, etc. Trace Genomics has developed an analytics engine to map the living soil and protect it.

It allows farmers to make better decisions about the crops that can be grown, their harvesting and future-planning for that land. By optimising planting of seeds, better seed-spacing and depth-control are guaranteed. Based on the amount of light reflected back to the optical crop sensors, sensing the correct amount of fertilisers needed by the crops is done in real time to prevent runoff of excess chemicals into nearby water bodies.

Variable-rate applications (VRA) are centred on understanding the relative productivity of different areas based on data obtained from sensors. Automated equipment and machinery can then be used to optimally apply herbicides, fertilisers and the like at variable rates and prevent overlapping. Maps and GPS ensure that machinery navigation and positioning are accurate. Nano-materials-based pesticides and insecticides further improve precision.

Smart weeding robot (Credit: https://en.reset.org)

Drones, both ground and aerial, are being used for crop planting and monitoring, field analysis, etc these days. With thermal or multispectral sensors, irrigation patterns essential for various fields can be identified. Irrigation can be controlled remotely through a PC or mobile. Benefits include integrated Geographic Information System (GIS) mapping as well as saving time and possibly increased yields. Several companies like DroneSeed, Agribotix, Skycision and HiveUAV have been providing drones for agricultural applications. Using drones, farmers can remotely manage crop operations without being a drone expert.

Utilising optical sensors, drones and satellite imagery, it is also possible to assess the health of various crops. Using mobiles and tablets, any pest population or weed activity on the farm can be checked, and then effectiveness of pest- and disease-control procedures can be evaluated. One of the most common methods is Normalized Difference Vegetation Index (NDVI). It uses wavelengths of infrared (IR) and visible light to detect crop variability and different crop stages.

Many companies are also building farm robots. These are commonly used for automation of agricultural processes like planting, irrigation, weed management, harvesting crops, fruit picking, ploughing, soil maintenance and so on. For instance, Ecorobotix and